

Subhan Shaikh

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SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, SQL.

AI / ML: LangChain, LangGraph, Llamaindex, RAG, MCP, LLM Fine-Tuning, LLM Evaluation, Claude (Anthropic) APIs

Agentic AI: Multi-agent systems & Orchestration, Computer Use Agents, Agent Guardrails & Evals, Claude Code, Codex

Backend & Data: Rest APIs, FastAPI, Node.js, React, Pydantic, PostgreSQL, Pinecone, ChromaDB, Redis, Supabase, Pgvector

Cloud & Deployment: AWS, Docker, Kubernetes, CI/CD, Github Actions

WORK EXPERIENCE

Gogentic.ai, Applied AI Engineer

June 2025 – Nov. 2025

- Led an AI and backend team of 6 people and key technical initiatives to deliver a prompt-to-3D-printing platform MVP in 8 weeks, enabling users to turn text, voice, or 2D images into custom 3D-printed parts.
- Built text, image, and audio-to-3D pipelines with retry and fallback logic across Meshy and self-hosted Hunyuan3D.
- Reduced infrastructure spend 80% by replacing paid Meshy APIs with Hunyuan3D in a self-hosted ComfyUI workflow, using custom image pre-processing to reach Meshy-comparable quality.
- Boosted backend performance 50% through quantized models, thumbnail storage in place of full 3D meshes and smart caching.
- Spearheaded post-launch customer discovery, converting direct user feedback into prioritized product improvements.

UC Santa Cruz - AIEA Lab, Graduate Researcher

June 2025 – March 2026

- Developed a UCSC CS department chatbot for internal teams, scraping and structuring department web data into course, academic, and graduation-requirement categories and implementing vanilla, hybrid, and agentic RAG pipelines for retrieval.
- Contributed to ongoing research on Prolog-based symbolic grounding to improve the factual precision and explainability of LLM outputs.

IBM, AI Intern

Sep 2022 – Nov 2022

- Built an end-to-end sign-language recognition system on the ASL dataset using OpenCV, MediaPipe, and TensorFlow, achieving 90% classification accuracy.
- Delivered a working prototype and presented a demo of the system and findings to the team.

PROJECTS

Articlr: LinkedIn Personal Branding Platform | articlr.com | LangGraph, LangChain, FastAPI, Supabase, Railway, Stripe

- Shipped a live SaaS used by real users, owning frontend, backend, auth, billing, and deployment end to end
- Designed a stateful LangGraph pipeline with three specialized agents, a Strategist that plans 3-5 distinct briefs, parallel Ghostwriters via asyncio, and a Critic with a capped revision loop that runs only on the selected draft, cutting critic calls ~80% per batch.
- Enforced output quality in code instead of prompts using Pydantic validated outputs, a banned-phrase and banned-structure filter that suppresses generic, templated LLM writing, and server-side overwriting of each draft's source citation so the model can't fabricate a credential.
- Built an ingestion pipeline on PyMuPDF with extraction-quality checks that soft-fails scanned PDFs, then distills each resume once into a structured digest reused by a 9-stage onboarding chain.

RepairMate : AI-Driven Home Device Diagnostics | [Github](#) | LangGraph, LangChain, OpenCV, Whisper, VLMs

- Designed and built a multi-agent, multimodal diagnostic platform that analyzes user-submitted repair videos by combining frame extraction, speech transcription, and Vision-Language reasoning to generate grounded troubleshooting reports and repair recommendations from appliance manuals, web resources, and instructional content.
- Developed an evaluation-driven optimization pipeline using GEPA to automatically refine frame-analysis and aggregation prompts against BERTScore-based objectives, improving diagnostic consistency and technical terminology accuracy. Applied model quantization and inference optimizations to reduce latency and enable real-time execution on resource-constrained devices.

Critique-Enhanced RAG: Self-Refining Retrieval-Augmented Generation | [GitHub](#) | Python, ChromaDB, OpenAI

- Built a self-critiquing RAG system that scores its own answers across four quality dimensions and conditionally refines those below threshold, improving oracle quality score 27% over a vanilla baseline on 150 SQuAD dataset questions.
- Benchmarked three architectures (Vanilla, LLM reranker, Critique-Refinement) on quality, cost, and latency, showing the critique loop refined 56% of answers and matched its gains at 2× cost while LLM reranking cost 6× for no measurable improvement.

EDUCATION

University of California, Santa Cruz

June 2026

Master's in Computer Science (GPA: 4.0)

Santa Cruz, CA

University of Mumbai

June 2024

Bachelor's in Computer Science

Mumbai, India